

Project Topic Examples

The following examples illustrate possible project directions under the five suggested areas. These are **only reference examples** to help inspire project ideas. Students are encouraged to design their own creative and technically interesting projects, especially those that can demonstrate a **live demo** during the poster session.

1. AI for Science

Using AI to assist scientific discovery, simulation, or scientific data analysis.

Example projects:

- Protein or molecule structure visualization and prediction tools
- AI-assisted drug discovery prototype
- Physics simulation predictors (e.g., predicting particle motion or physical dynamics)
- Scientific data analysis assistant (for astronomy, biology, or materials science datasets)
- AI-based material property prediction systems

Possible demo:

- Interactive molecule visualization
- Real-time prediction of simulation results
- Scientific dataset exploration interface

2. AI + X (Domain Applications)

Applying AI technologies to real-world domains such as education, healthcare, finance, or software engineering.

Example projects:

- AI-powered coding assistant or debugging helper
- AI-based learning tutor or homework explanation system
- Intelligent financial analysis dashboard
- AI tool for design ideation or UI generation

Possible demo:

- Real-time coding assistant
- Interactive tutoring chatbot
- AI-assisted analysis dashboard

3. AIGC (AI Generated Content)

Systems that generate creative or structured content such as text, images, audio, or videos.

Example projects:

- AI story or novel generator
- Image generation tool with style control
- AI music or sound generation prototype
- Prompt-based poster or logo generator

Possible demo:

- Real-time image generation
- Interactive creative writing assistant
- Prompt-based design generation tool

4. Multi-modal AI

AI systems that process and combine multiple modalities such as text, images, audio, or video.

Example projects:

- Image captioning systems
- Visual question answering systems
- Multimodal chatbot (image + text interaction)
- Video understanding and summarization tools

Possible demo:

- Upload an image and ask questions about it
- Interactive multimodal assistant

5. AI Agents

Autonomous AI systems that can plan, use tools, and perform multi-step tasks.

Example projects:

- AI agent that browses websites and summarizes information
- Multi-agent collaboration systems
- Task automation agents (e.g., planning daily schedules or workflows)
- Tool-using LLM agents integrating APIs and external tools

Possible demo:

- Agent completing a multi-step task automatically
- Multi-agent simulation environment
- Interactive AI assistant performing real tasks

6. Physical AI

Developing embodied AI systems by combining robotic perception, 3D world modeling, and learning-based control (e.g., reinforcement learning, imitation learning, and foundation models).

Example projects:

- Imitation learning models using behavior cloning
- Implementing reinforcement learning for low-level control
- Evaluating or adapting vision-language-action robot foundation models
- 3D reconstruction pipelines using implicit fields like NeRF or SDF
- World Model based video generation or reconstruction

Possible demo:

- Simulation environment demonstrating a learned RL policy controlling a robot
- Interactive 3D reconstruction visualization from scene inputs
- Interactive robot policy setup showcasing behavior cloning

These examples are meant to **provide inspiration rather than strict limitations**.